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Face Recognition using Geometric Measurements, Directional Edges and Directional Multiresolution Information

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Abstract

This paper propose the extraction of geometric measurements of face regions, different directional information and multiscale information offered by real tree dual discrete wavelet transform to generate facial features for face recognition. The proposed scheme calculates feature sets: geometric measurements such as the area measures of eye, nose and mouth, horizontal, vertical, and diagonal directional edge information and multiscale information using whole face information. Existing feature-based or local features based methods, depends on characterization of individual facial features (i.e., eyes, nose, and mouth etc.) and their geometrical relationships. Our approach automatically calculates the measures of important areas of interest that describe the information present in an individual face image. The area of different segmented regions is calculated that describes the information of facial components. The proposed geometric features, directional information using Fast Wavelet transform and multiscale image formation gives the rotation, pose and expression invariant face recognition. Fusion of various generated features improves the recognition considering the different segmented face information. The success of the scheme is the automatic calculation of various geometric features, directional edge information and whole face information. The main features of our approach are detection of important regions, calculation of geometric features and fusion with information offered by edge and multiscale information. We demonstrate the success of our approach by experiments.

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Keywords: Geometric measurements; Fast Wavelat Transform directional edge; Multiscale face information

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1. Introduction

In Face recognition we recognize an individual or verify a person claimed identity using human face and its individual regions. Nowadays segmented facial regions, such as forehead, eyes, nose, mouth, chin and the distance between them attracted attentions of researchers. Features extracted from the individual regions and the distance between these found to be unique for an individual. The facial measurements remain to be constant once a person get mature and is slightly affected even if the face is rotated or tilt playing a role in face recognition.

The idea is to segment the individual regions and get the distinctive features for an individual. Calculating exact distance and relative positions is a great challenge in image processing giving false recognition even though there is vast difference between the individual regions. Holistic feature generation technique contains less discrimination details whereas a geometrical measurement founds to be more discriminate for an individual. Face features can be extracted either automatically or manually from the individual regions and then it is given to system which performs the recognition. Face recognition can successfully recognize faces, only when images are obtained under certain prescribed conditions. The performance of the system degrades when face images are captured under varying pose, lighting, with accessories and expression. The image variations caused by the illumination conditions, viewing conditions or poses, different facial expression, ageing, and disguises such as slight cut, glasses or makeup should not affect the recognition.

In this paper we propose a new approach using geometric measurements of face regions such as eyes, nose, and mouth. Directional information is calculated using Fast Wavelet Transform. Whole face information is generated using Real Dual Tree Discrete Wavelet Transform (DT-DWT) for robustness of multi resolution and multi directional approach for pose and expression invariant face recognition. Fusion is done using magnitude of detailed subbands at each level of real Dual tree discrete wavelet transform (DT-DWT). In order to capture better multi-orientation information in face images, each face image is described by a subset of band filtered images containing coefficients of DT-DWT which characterize the face textures. The overall texture features of an image at each resolution are obtained through fusion of the detailed subbands. We fused subbands into small sub-blocks, from which we extract compact and meaningful feature vectors using statistical measures such as feature variance. Feature variance is calculated using the features measure such as mean and standard deviation.In this paper we calculate geometric measurements (area) of face regions, directional information and then we fuse it with the Real Dual Tree Discrete Wavelet Transform. Selesnick's Real Dual-Tree 2D wavelet transform is implemented using two critically sampled DWTs in parallel on the same image. When compared the real Dual tree wavelet transform with Gabor wavelets it is found that Gabor wavelets are relatively computational expensive. The wavelets used are a Hilbert pair if the halfband filter has a fractional-delay allpass factor.

The rest of the paper is organized as follows. In section 2, we introduced the related work describing the current state of the art. In section 3, proposed technique is described in detail. In Section 4, experimental results and comparisons are explained. Finally, conclusions are given in Section 5.

2. Related Work

Face recognition excellent survey was given by Zhao et al., Chellappa et al., describe an approach for holistic feature. Turk, M, and Pentland use principal components analysis by defining Eigenface for face recognition. Nefian, and Hayes, created Hidden Markov Model for face recognition. Wiskott et al., and Xu, et al., describes an excellent work that based on component and morphable models. Jiang and Feng used DCT to calculate facial features. Shen et al., and Nicholl, and Amira used the properties of discrete wavelet transform (DWT) in recognizing an individual. Multiresolution analysis using 2D wavelet transform extract multiple

subband face images. Multiresolution methods provide powerful signal analysis tools, which are widely used in feature extraction, image compression and denoising applications. Wavelet decomposition is the most widely used Multiresolution technique in image processing and for face recognition to overcome the problems of face appearances, orientation, expression, accessories. DWT provides an "efficient" basis representation of regular signals. DWT limitations are the aliasing effects in transform domain, oscillations of the coefficients around singularities and a lack of shift invariance. Al-Amin Bhuiyan, Chengjun Liu and Harryechsler, Shen et al., use Gabor wavelets in face recognition. It is computationally very complex and the memory requirements for storing Gabor features are very high. The dual-tree complex wavelet transform are good for holding directional multiresolution with excellent directionality, reduced shift sensitivity and explicit phase information. N. Kingsbury proposed the Dual-tree Discrete Wavelet transform that again investigated by I. Selesnick in the dyadic case. I. Selesnick, R. Baraniuk and N. Kingsbury provided an overview of the topic with an application. Ekenel and Sankur, use the property of wavelet multiresolution to cope up with noise and non-uniform illumination in generating the facial feature. The real Dual tree wavelet transform by I. Selesnick gives directional multiresolution decomposition of a given image much like the Dual tree complex wavelet transform and Gabor wavelets and is relatively less computational expensive. Hazim et al., used Multiresolution property for face Recognition. Fast Wavelet Transform is directionality sensitivity and is useful in detecting edges helps in capturing directional edges in face recognition given by R. C. Gonzalez, R. E. Woods, & S. L. Eddins.

Kanade proposes the automatic recognition of faces by generating a vector using geometric features. He locates facial feature points, such as eye corners, nose, mouth, chin counter. The template matching strategy has the advantage of has speed correlation with restriction of being sensitive to translation, size, illumination and orientation. Roberto Brunelli and Tomaso Poggio developed two algorithms; the first algorithm is based on the computation from a set of geometric features calculated by considering nose width and length, the position of the mouth, the shape of the chin. The second algorithm is based on the strategy of template matching by considering the information of gray level image data. Young-Jun Song et al., extracted the geometric features and the technique is validated using Principle Component Analysis/Linear Discriminant Analysis. The facial regions are divided according to the position of the eye, nose and mouth. Discrete Wavelet transform at level 3 is used for capturing the Whole face information.

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3. Proposed Work

The proposed technique is a five step procedure; (1) In first step we convert gray image to black and white (2) In second step segmentation of facial regions such as eyes, nose and mouth is done. (3) In the third step the areas of different face regions is calculated. (4) In the fourth step directional information is calculated using Fast Wavelet Transform (5) In the fifth step Selesnick's toolbox is used for calculating (level two) Real Dual Tree Discrete Wavelet Transform so as to capture the variations of pose, expression, orientations. A (DT-DWT) produces two band-pass sub-images at each level, corresponding to lo-hi, hi-hi and hi-lo filtering. After this we perform normalization is done with the help of feature variance. (6) Finally the fusion of the data is performed that is generated from the all above steps.

3.1. Binary Image Formation

The gray level image contains the gray level information ranging from 0-255. The conversion of the gray level image to black and white is done by converting all the pixels of the image with the luminance greater than the threshold. Otsu, algorithm set the threshold for better reconstruction to transform the image. The

Otsu's method chooses the threshold which minimizes the intraclass variance of the black and white pixels. Black and White image formation converts the `gray level image to` a normalized intensity value that lies in the range `[0, 1]`. The gray level values greater than the threshold given by Otsu's method is set to 1 and the gray level values less than the threshold are set to 0. For example, if the image is of size 112X92, the Otsu's method threshold is found to be 0.4627 for the input image as shown in Fig.1

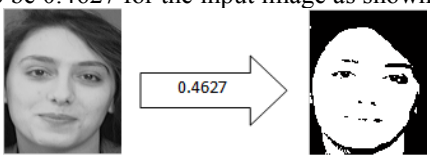

Fig. 1. (a) Binary image

3.2. Segmentation of facial Regions

Eyes constitute an important region of the face. Careful investigation suggests that eyes are nearly unique for each individual. Also there may be a possibility that there exists a difference between left and right eyes. Proposed approach segments the left and right eye of an individual to capture the meaningful information. Eyes region are segmented after several analysis and is found to be by selecting the rows from 31:55 and columns from 55:78 for the right eye region. For the left eye region rows from 31:55 and columns from 15: 40 are selected. Mouth is an important organ of face and is found to be nearly unique for each individual. It is segmented by considering the rows from 78:96 and columns from 31:70. Nose is also an important organ of face and is found to be unique for an individual. It is segmented by considering the rows from 48:75 and columns from 38:62. Some more regions near to eyes are selected so as to make the system invariant to expression, orientation and profile.

Fig. 2 shows the different segmented region of the face for the binary transformed image. Segmentation is done on the black and white image that we generated as explained in section 3.1 different segmented facial regions are unique in the sense that their width, length and area differ from person to person.

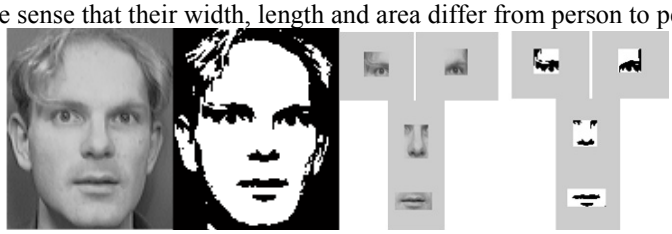

Fig. 2. Segmented regions of face

3.3. Area of Facial Segmented Regions

After segmenting the different facial regions as proposed in section 3.2, the areas of the different segmented regions are computed as follows:

The area is calculated based on roughly the total number of on pixels in the image, but they may not be exactly the same because different patterns of pixels are weighted differently. Each pixel is part of four different 2-by-2 neighbourhoods. The area of an each pixel is calculated by gathering information based on 2-by-2 neighbourhood using bit Quad Patterns as explained by Pratt and William.

Fig. 3. (a) Face Image (b) Left eye (0.0580); (c) Right eye (0.0523); (d) Nose (0.0806) (e) Mouth (0.0869)

Fig.3 shows segmented facial regions along with its corresponding area. Normalization of the different areas is done by dividing it with 7000 so as get the values in required range.

3.4. Fast Wavelet Transform

Fast Wavelet Transform is directionality sensitivity and its usefulness in edge detection helps in capturing directional edges in face recognition. It is defined as the linear combination of double resolution with copies of themselves via series of expansion. To make the system direction and expression invariant, facial features are extracted using Fast Wavelet transform. It gives in detail the horizontal, vertical and diagonal coefficients. It is performed using wavelet 'sym4'. The detail coefficients are retrieved by convolving rows of image with high pass filter and then downsampling. The columns are then convolved with high pass and low pass filter to get the diagonal and vertical coefficients. The rows are convolved with the low pass filter followed by downsampling. Finally the columns are get convolved with high pass and low pass filter to get horizontal and approximation coefficients. It is almost twice fast as when compared with Discrete Wavelet Transform.

Fig. 4 shows the result of Fast Wavelet Transform of the face image after convolving with the wavelet. Horizontal coefficients provide the information of facial details such as Eyes, nose, mouth and ear. Vertical coefficients provide the information of the face structure.

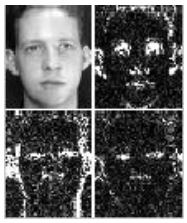

Fig. 4. (a) Face image along with Horizontal, Vertical and Diagonal information

3.5. Real Dual Tree Discrete Wavelet Transform Image Formation

In 2D DWT we get 3 detail subbands excluding the approximation subband at each level. First two wavelets in 2D DWT are oriented in the vertical and horizontal directions whereas the third wavelet does not have a dominant orientation. The third wavelet mixes two diagonal orientations giving rise to the checkerboard artifact. i.e., the 2D DWT is poor at isolating the two diagonal orientations. In Real DT-DWT each subband contains Dual tree (2 tree) structure to form two different orientations. In Real DT-DWT the six wavelet are oriented in six distinct directions and free of checker board artifact as shown in Fig.5.

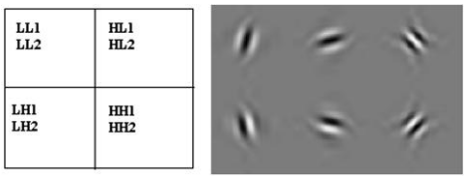

Fig. 5. DT-DWT along with its subband

In this step we acquire directional multiscale decomposition of the gray level whole face image up to level 2. Real Dual Tree Discrete Wavelet Transform (Selesnick's technique) is used in order to make the system capable for holding various expressions, and to cope up with the problems of orientation, illumination and pose. A (DT-DWT) produces two band-pass sub-images at each level, corresponding to lo-hi, hi-hi and hi-lo filtering. For an face image of size [128x128], six band-pass sub images of size [64x64] and eight band-pass sub images of size [32x32] giving total of 14 images generated using real DT-DWT (level 2) are generated.

The features from 14 band-pass sub images are extracted using the statistical measure. Feature variance is used to normalize the data and to get the facial description. It is calculated as follows:

- Generate the face image vector from the training dataset by reshaping each band image to 64x64 and then stacking.
- Let the generated reshaped image vectors face images be
  $\Gamma_1, \Gamma_2, \Gamma_3, \ldots, \Gamma_N$
- Calculate the mean image from the training set, and is defined by
  $$\Psi = \frac{1}{N}\sum_{n=1}^{N}\Gamma_n .$$
- Each face differs from the average by the vector
  $$\Phi_n = \Gamma_n - \Psi$$
- Calculate the standard deviation as
  $$std = \left(\frac{1}{N-1}\sum_{i=1}^{N}(\Gamma_i - mean(\Gamma_i))^2\right)^{1/2}$$
- Finally generate Feature vector by dividing each vector of Φ_n with standard deviation of corresponding image vector.

3.6. Fusion

Facial description vector is generated from the multiscale description information of the whole face image data as explained in step 3.5. Finally fusion is performed by stacking the vector with the geometric normalized measurements explained in step 3.3, Fast Wavelet Transform edge gradient information as explained in step 3.4 and facial vector as explained in step 3.5.

4. Experimental Results

The proposed technique is implemented and validated in MATLAB 2010. As the proposed work is based on the information gathering from geometric measurements as well as using the information generated from whole face information, we form the training set consisting of 3 pose per person where the images are mainly

frontal face images with some variations. The training dataset for the proposed work contains Frontal images with different lighting conditions and the test dataset contains images that are different than the training dataset. The training and the test dataset generated using the Olivetti Research Laboratory datasets (ORL) at the AT & T Laboratories Cambridge lab given by Ferdinando et al.,

Recognition is done on the using the proposed approach and Mahalanobis distance measure is used as the distance measure as explained by Shiming Xiang et al., All the input test images were correctly recognized from the training dataset even if there is change in emotions like- neutral, smile, laughter, sad, slight change in poses like looking front, left, right, up, or looking down. The accessory does not affect the recognition of the system

Table 1. Shows the comparisons of the proposed technique with the technique given by Young-Jun Song et al., use wavelet transform and capture geometric features followed by the PCA/LDA approach. The Eigenface given by Turk, M., and Pentland is also compared. Struc and Pavesiae DCT normalization technique is also compared after applying PCA.

Table 1. Recognition Rate with its approaches

Approach and Features	Recognition Rate
Young-Jun Song et al., PCA/LDA + Geometric information (Wavelet Features)	94 %
Turk, M., and Pentland, PCA(Eigenface Features)	86 %
Struc and Pavesiae, PCA(DCT Normalization features)	86 %
Proposed (Geometric measurements fusing with RDT-DWT)	**95 %**

5. Conclusion

The existing techniques capture the different segmented geometric measurements, edge directional information and whole face information. The merging of the captured information is performed to generate feature vector. To discriminate the data, feature variance and normalization are applied. The proposed work handles the various issues of face recognition such as different expressions, variations in pose, orientation, accessories and makeup. Multiscale descriptions of face have advantage over the DT-CWT in terms of computational efficiency. The generated normalized feature vector has the capability to discriminate the individual without being operated on PCA/LDA claiming in saving of time and space. In the future more number of facial measurements will be added with the consideration of orientation and illumination problem. The comparison of the proposed work is carried out with the existing techniques used to improve the effectiveness of the proposed technique.

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